

**Project: Generative AI-Based Ligand Binding Pose Refinement from Physics-Based Priors**

**Report 1: Understanding Proteins, Protein-Ligand Interactions and Molecular Docking**

Topics:

- 1) Proteins (Amino Acids)
- 2) Protein-Ligand Interaction
- 3) Molecular Docking

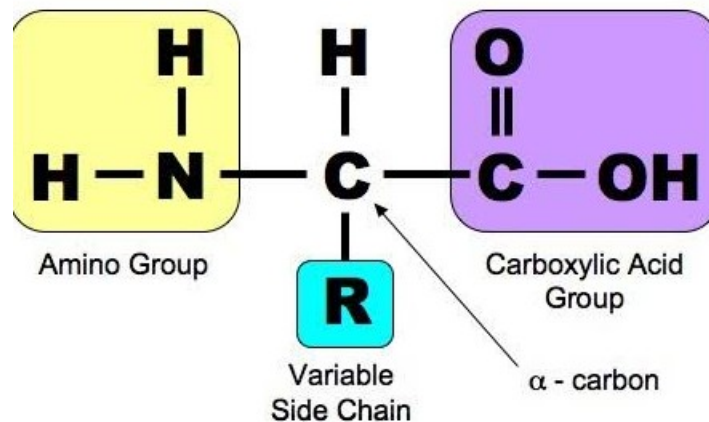
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## **I PROTEINS:**

- (i) Proteins transport nutrients, catalyze chemical reactions and are the structural components of living organisms.
- (ii) Proteins are made up of 20 set of standard amino acids.

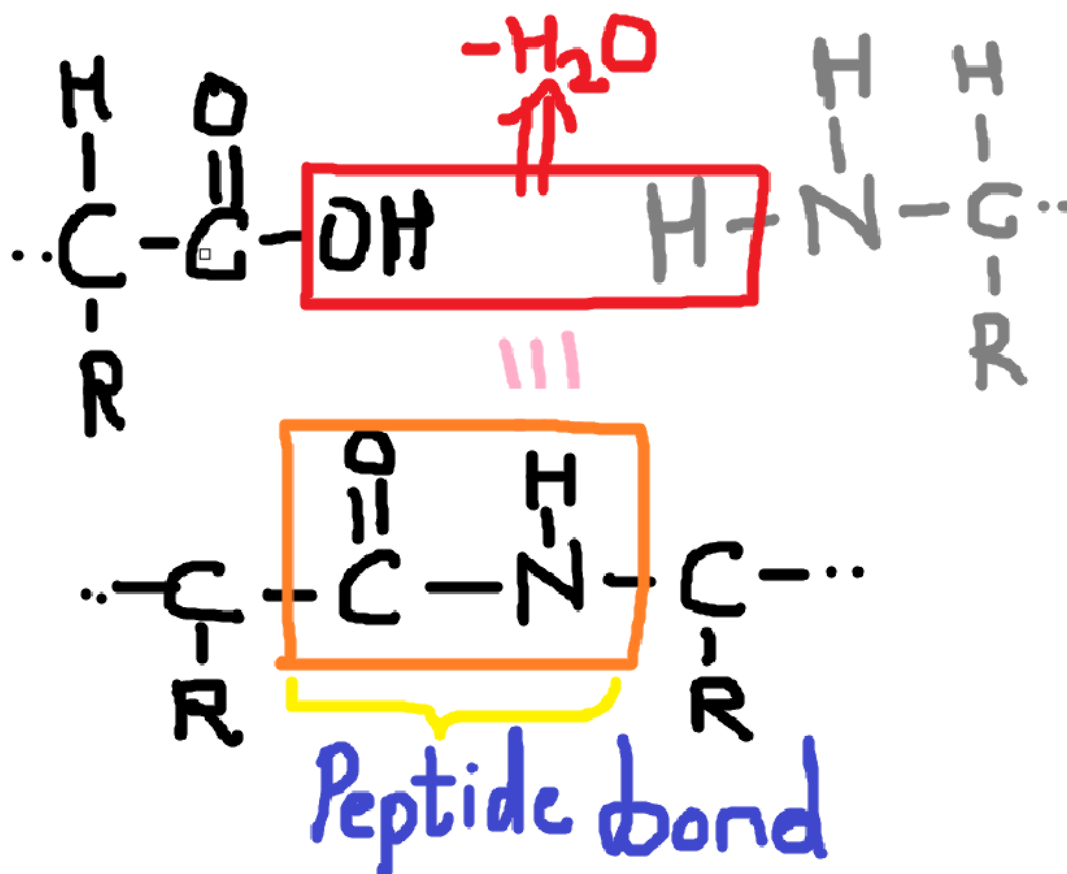
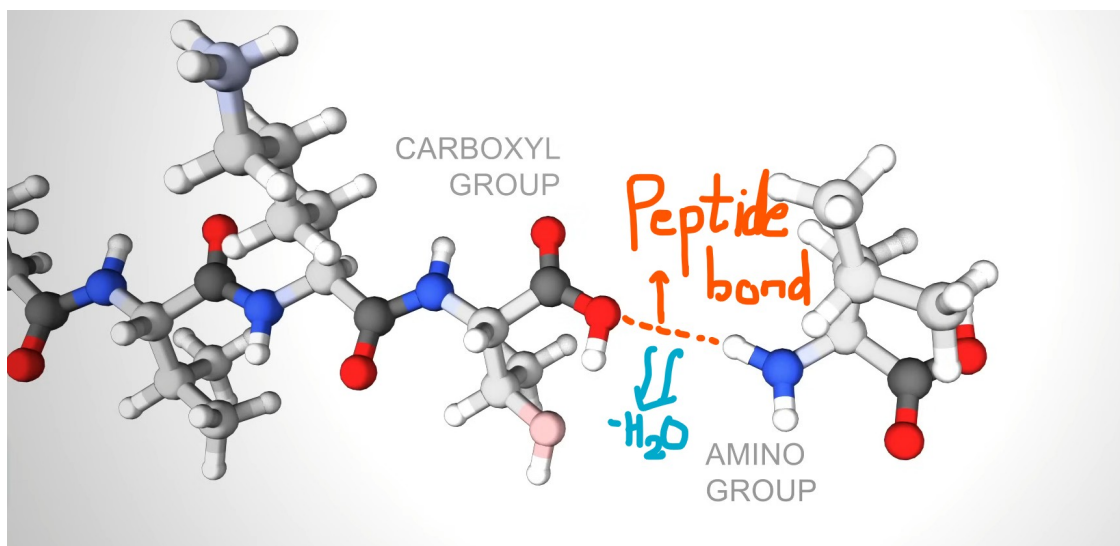
### **AMINO ACIDS:**



- (i) Amino Acids are classified by their side chain (-R group) structure.
  - a. Hydrophobic: The side chain that repel water molecules. (non-polar group).
  - b. Hydrophilic: The side chain which favorably interact with water molecules (through hydrogen bonding).
  - c. Charged: Interact with oppositely charged molecules/amino acids. (acidic or basic groups).

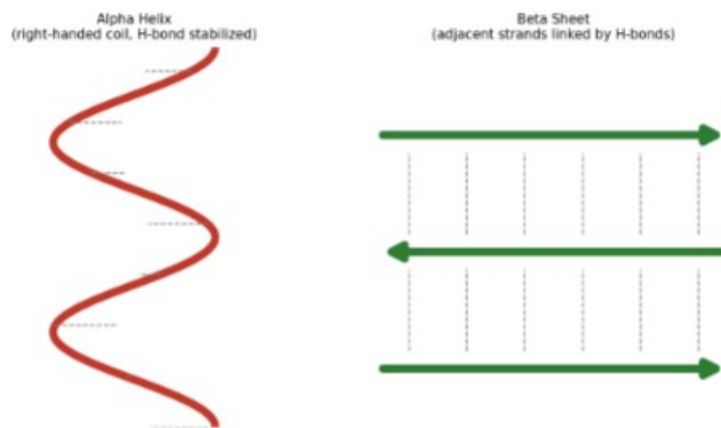
### **PRIMARY STRUCTURE OF PROTEINS:**

- (i) The primary structure of proteins is formed due to the linear sequential joining of amino acids as encoded by the DNA.
- (ii) The amino acids are joined via peptide bonds.



## **SECONDARY STRUCTURE OF PROTEINS:**

- (i) Alpha helix: It's a right-handed coil stabilized by hydrogen bonds between nearby amine and carboxylic groups.
- (ii) Beta sheet: It is formed when hydrogen bonds stabilize two or more adjacent strands.



## **TERTIARY STRUCTURE OF PROTEINS:**

The tertiary structure is the overall 3D shape of a single protein chain, determined by the properties of its amino acids.

- (i) Many proteins fold into globular shapes with hydrophobic side chains tucked inside, shielded from surrounding water.
- (ii) Membrane-bound proteins instead cluster hydrophobic residues on the outside, allowing them to interact with lipid membranes.
- (iii) Charged amino acids let proteins interact with molecules bearing complementary (opposite) charges.

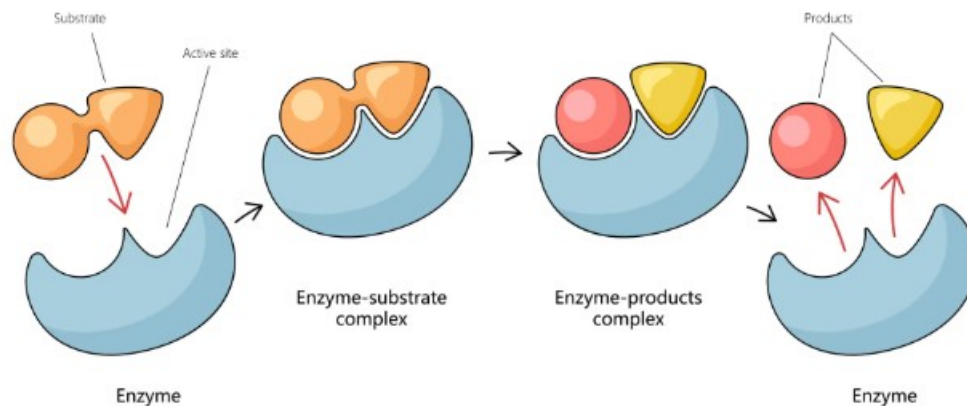
## **II PROTEIN-LIGAND INTERACTION**

### **ENZYME-SUBSTRATE INTERACTIONS:**

- i. Enzymes are special proteins with a specific part typically referred to as the “*lock*” (active site).
- ii. Substrates are the “*keys*” (molecules) that fit into the *lock*.
- iii. When a substrate binds to an enzyme, it undergoes a chemical reaction and is converted into a product.
- iv. Enzyme releases the product after the reaction and returns to its original shape.

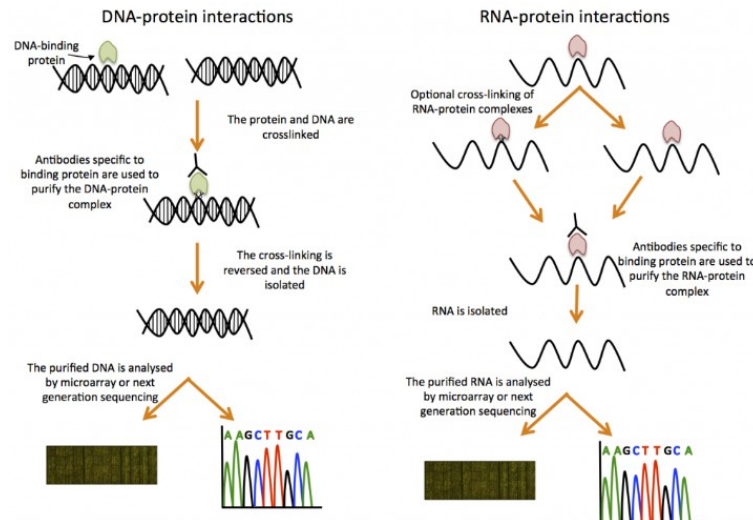
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### Mechanism of Enzyme Action



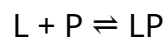
### **PROTEIN-NUCLEIC ACID INTERACTIONS:**

- i. Proteins can connect to nucleic acids (DNA and RNA) to regulate gene expression, DNA replication, and RNA processing.
- ii. Some proteins can bind to specific spots in the DNA and prevent those spots from being “read”, ensuring that genes are activated or suppressed when needed.



## **BINDING AFFINITY AND DISSOCIATION CONSTANT:**

- Binding affinity is the measure of how tightly a ligand binds with the active site.
- Suppose we represent ligand as {L} and protein as {P} then when they are bound together we will represent them as {LP}. This is a two-state process and can be represented as:



- The dissociation constant can be calculated as:

$$k_D = \frac{[L][P]}{[LP]}$$

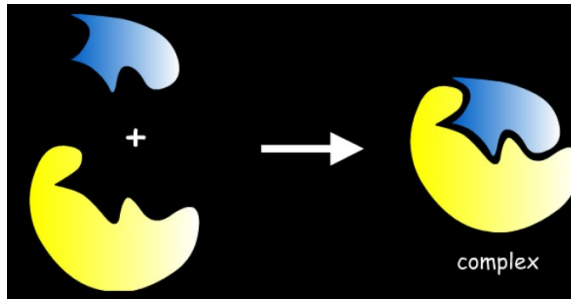
- A lower dissociation constant means higher binding affinity and a higher dissociation constant means lower binding affinity.
- Ideal drug candidates have higher binding affinity to intended targets.

## **III MOLECULAR DOCKING**

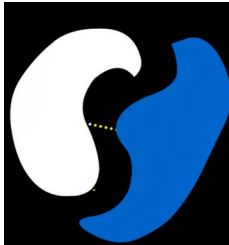
### **UNDERSTANDING DOCKING**

- Molecular recognition is the ability of biomolecules to recognize other biomolecules and selectively interact with them in order to promote fundamental biological events this is known as Molecular Docking.
- Types of Molecular Docking:

- a. Lock and Key Theory: A substrate fits into the active site of a macromolecule, just like a key fits into a lock.



- b. Induced-Fit Theory: Both ligand and target mutually adapt their structure until an optimal fit is achieved.

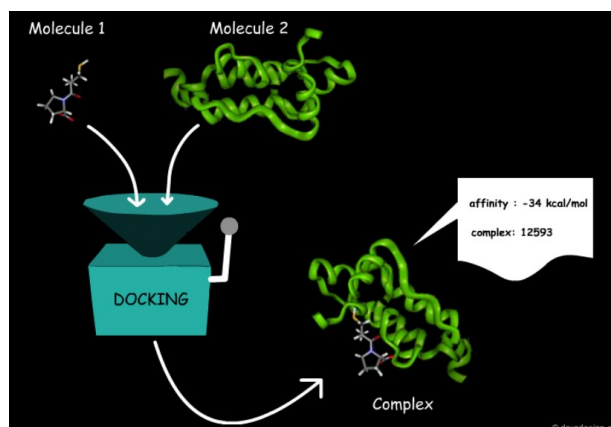


### **COMPUTATIONAL DOCKING:**

- i. It is a computational science aiming at predicting the optimal binding pose of interacting molecules.
- ii. Most common docking systems: protein-protein, nucleic acid-protein and protein-small molecule.

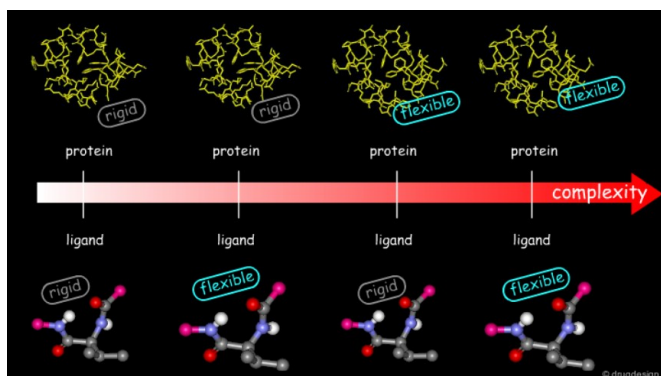
### **DOCKING PROBLEM:**

- i. The task of predicting how molecules interact with each other is called docking problem.
- ii. Molecular docking software is like a black box whose input is set of 3D structures to be docked and output is the predicted complexes of molecules.
- iii. The main goal is to find the 3D structure of the complex and binding affinity.



## CLASSIFICATION OF DOCKING PROBLEMS

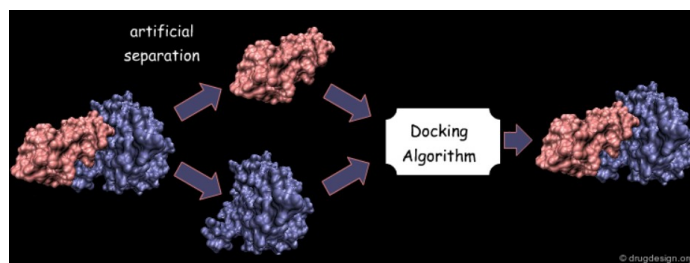
By Molecular Flexibility:



By Availability of Structural Data:

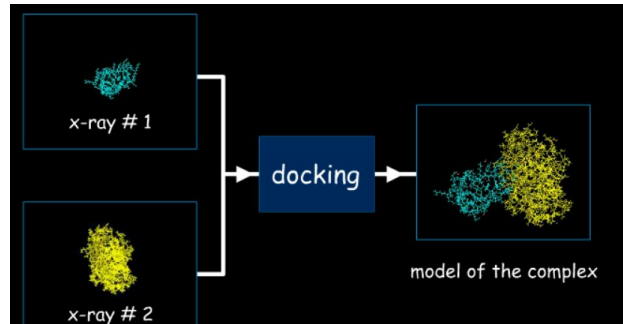
Docking difficulty also depends on what structural information is available in the start of the process.

- i. Bound Docking: The experimental structure of complex is already known. Used mainly to calibrate the software.

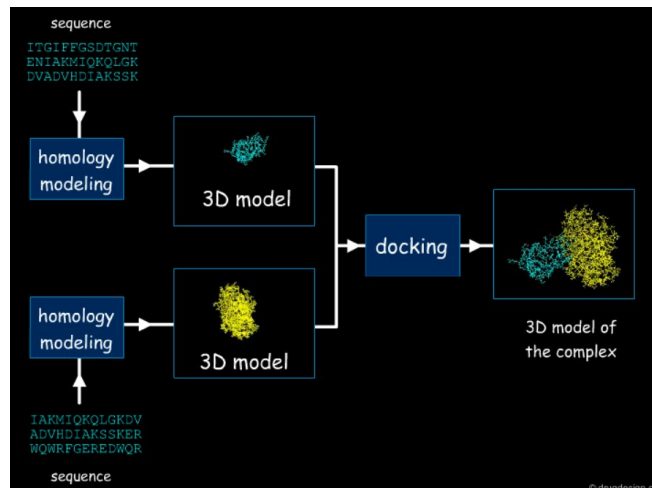




- ii. Unbound Docking: Only the separate molecule structures are known. You'll have to correctly develop the model for this condition.



- iii. Modeled Docking: No experimental structure at all. It relies on homology modeling.



## SCORING METHODS

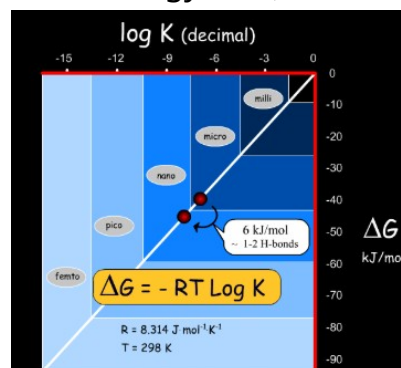
- i. Binding is governed by three main factors: interaction energies, desolvation/solvation energies and entropic effects.
- ii. Interaction Forces:

- a. Favorable Forces: Good hydrophobic contacts, hydrogen bonding (strong and directional), attractive electrostatic (strong and directional).
- b. Unfavorable Forces: bad hydrophobic contacts (steric clashes) and repulsive electrostatic interactions.
- iii. Desolvation: Hydrophobic groups associate to reduce contact with surrounding water.
- iv. Entropy: Molecular flexibility affects binding energy.

## **BINDING AND FREE ENERGY**

- i. Binding energy is the energy required to separate complex into its parts;  

$$= E(\text{complex}) - [E(\text{protein}) + E(\text{ligand})]$$
- ii.  $\Delta G = -RT \ln(K)$
- iii.  $\Delta G$  change of about 6 kJ/mol corresponds to roughly one order of magnitude in log K - which can be the difference between a micromolar and a nanomolar compound.
- iv. Calculating  $\Delta G$  directly from the Gibbs equation is hard because the entropic term is difficult to estimate, so it's often neglected as a first approximation.
- v. With entropy neglected and constant pressure assumed,  $\Delta G$  can be approximated using internal energy ( $\Delta U$ ) via molecular mechanics.



## **SCORING FUNCTION**

- i. Two categories:

- a. Empirical scoring function: The software counts how many specific interactions occur in a modeled drug-protein pose; Each type of interaction is multiplied by a predefined weight.

$$E_{\text{Total}} = k_{\text{VdW}} \cdot E_{\text{VdW}} + k_{\text{es}} \cdot E_{\text{es}} + k_{\text{hb}} \cdot E_{\text{hb}} + \dots \dots \dots + k_{\text{torsional}} \cdot E_{\text{torsional}} + k_{\text{desolvation}} \cdot E_{\text{desolvation}}$$

- b. Knowledge-Based scoring function: Computational tools scan the database (PDB); The tools count how many times atom pair occur at some specific distance; Using statistical mechanics, high-frequency distances are assumed to be energetically favourable (low energy), while rare distances are classified as unfavourable (high energy); These probabilities are converted into statistical energy scores called PMFs.